Power BI - Introduction

Power BI is a data visualization and business intelligence tool developed by Microsoft. It helps users connect to, transform, analyze and visualize data through interactive reports and dashboards.

- **Interactive Reporting:** Creates dynamic reports and dashboards that allow users to explore data.
- **Data Analysis:** Helps identify trends, patterns and relationships in data.
- **Data Integration:** Brings data from multiple sources into a single environment for analysis.
- **Business Intelligence:** Turns data into actionable insights that support business operations and planning.

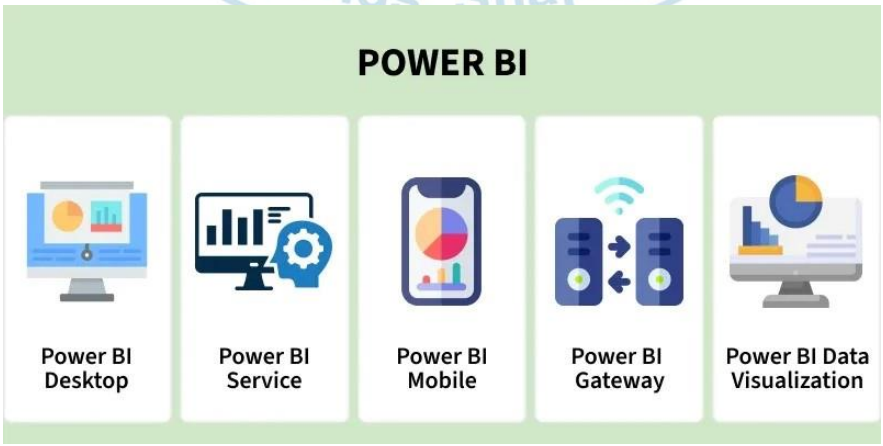
Need for Power BI

Organizations often work with large amounts of data spread across different sources, making it difficult to analyze and monitor information efficiently. Power BI addresses these challenges by providing a centralized platform for data analysis and reporting.

- **Centralized Analysis:** Combines data from different sources into a single view for easier analysis.
- **Faster Reporting:** Reduces manual effort involved in preparing and updating reports.
- **Data Monitoring:** Helps track changing data and key metrics through regularly refreshed or real-time reports.
- **Self-Service Analytics:** Allows users to explore data and create reports without extensive programming knowledge.
- **Scalable Reporting:** Supports individual users as well as organizations with large-scale reporting and data needs.

Key Components

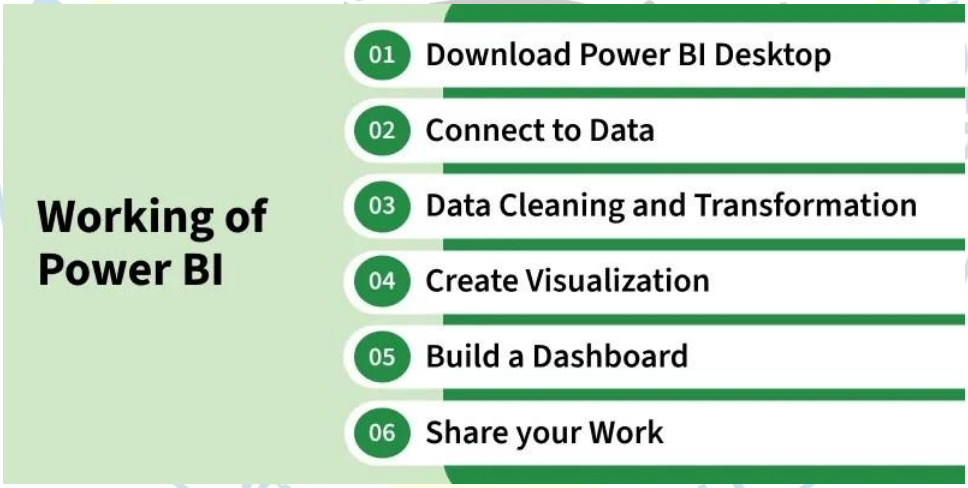
Power BI consists of several components, each serving a specific role in the data analysis and reporting process:



- **Power BI Desktop:** A free application installed on your computer that allows you to connect to data, clean it and create interactive reports.
- **Power BI Service:** The cloud-based version where you can publish, share and access reports and dashboards from anywhere.
- **Power BI Mobile:** Mobile apps for iOS and Android that let you view and interact with reports on smartphones and tablets.
- **Power BI Gateway:** Acts as a bridge between on-premises data sources and Microsoft cloud services such as Power BI, allowing data to be refreshed or accessed securely.
- **Power BI Report Server:** An on-premises reporting platform that allows organizations to publish and manage Power BI reports within their own infrastructure.
- **Power BI Embedded:** Allows developers to embed Power BI analytics and reports into applications and websites, enabling users to access insights within the application.

How to Use Power BI

Getting started with Power BI is simple. Follow these steps to create insightful reports and dashboards.



Step 1: Download Power BI Desktop

- Download Power BI Desktop for free from the Microsoft website.
- Install it on your Windows system to start creating reports.
- Provides a clean and easy-to-use workspace.

Step 2: Connect to Your Data

- Click Get Data to connect to Excel, SQL databases or cloud sources.
- Import data from multiple sources into one report.

- Follow simple on-screen steps to load your data.

Step 3: Clean & Transform Data

- Use Power Query to clean and prepare your data.
- Remove unnecessary data and fix data types.
- Ensure accurate and reliable analysis.

Step 4: Build the Data Model

- Create relationships between related tables.
- Organize tables into a suitable data model.
- Add calculated columns and measures when required.
- Ensure the model is structured for efficient analysis.

Step 5: Create Visualizations & Reports

- Drag and drop fields to create charts, tables, cards and other visuals.
- Use filters and slicers to make reports interactive.
- Arrange visuals to present important metrics and insights clearly.

Step 6: Publish and Share

- Publish the report to Power BI Service.
- Share reports or dashboards with users based on permissions.
- Configure scheduled refresh when required.

How to Install Power BI on Windows?

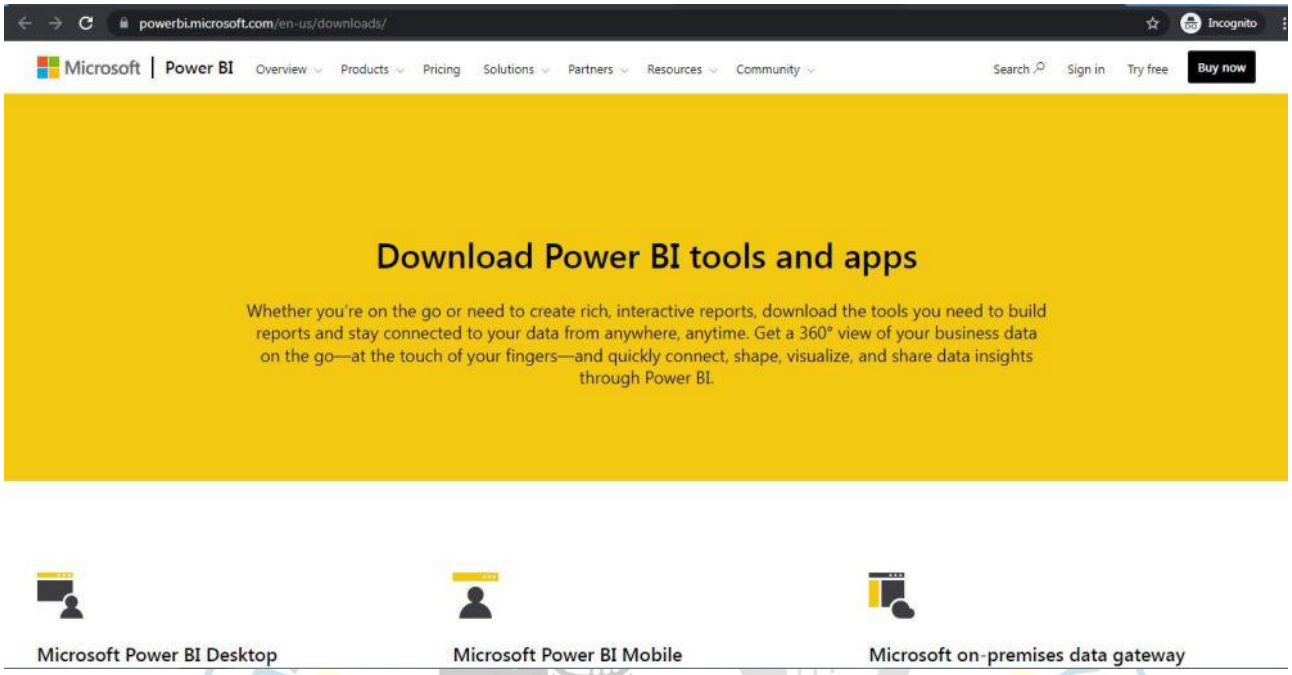
- **Power BI** is a business analytics tool developed by Microsoft where "BI" stands for Business Intelligence. It is used to connect, visualize and analyze data from various sources. You can create interactive dashboards, reports, graphs and charts with just a few clicks. Power BI also helps with data cleaning, preparation and discovery.

Steps to Install Power BI on Windows

Follow these simple steps to install Power BI on your Windows computer:

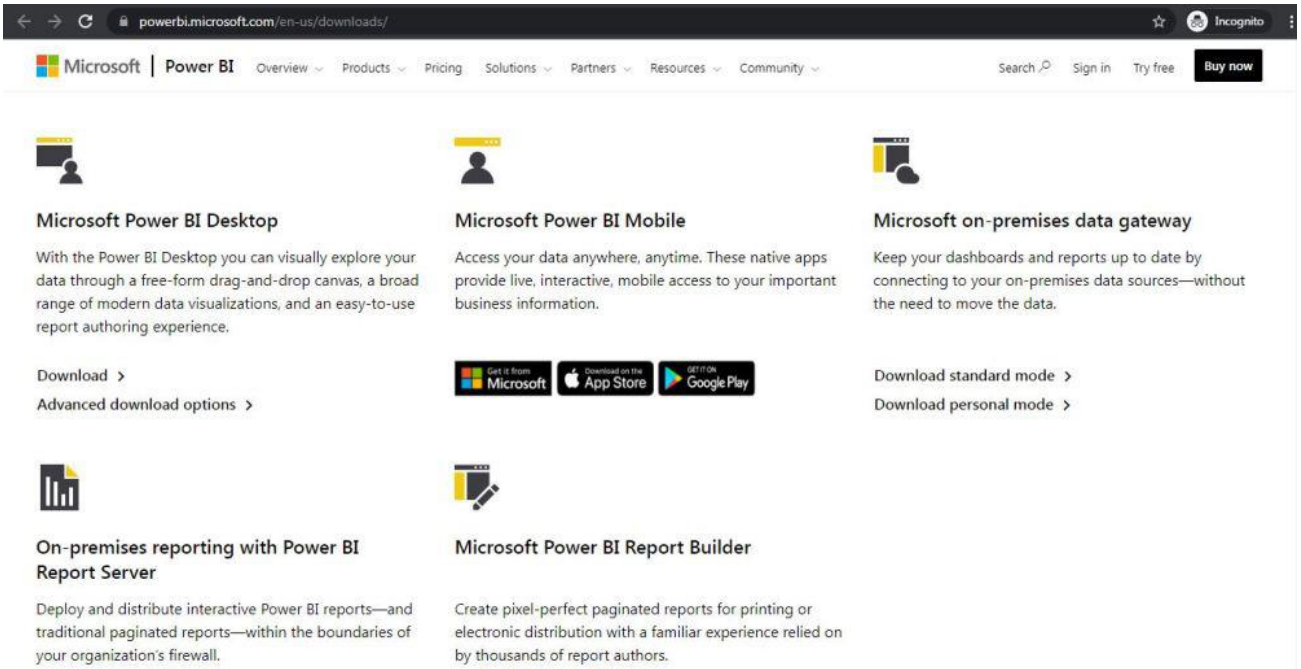
Step 1: Open the Official Power BI Website

Open any web browser like Chrome or Edge and go to the official Power BI website.



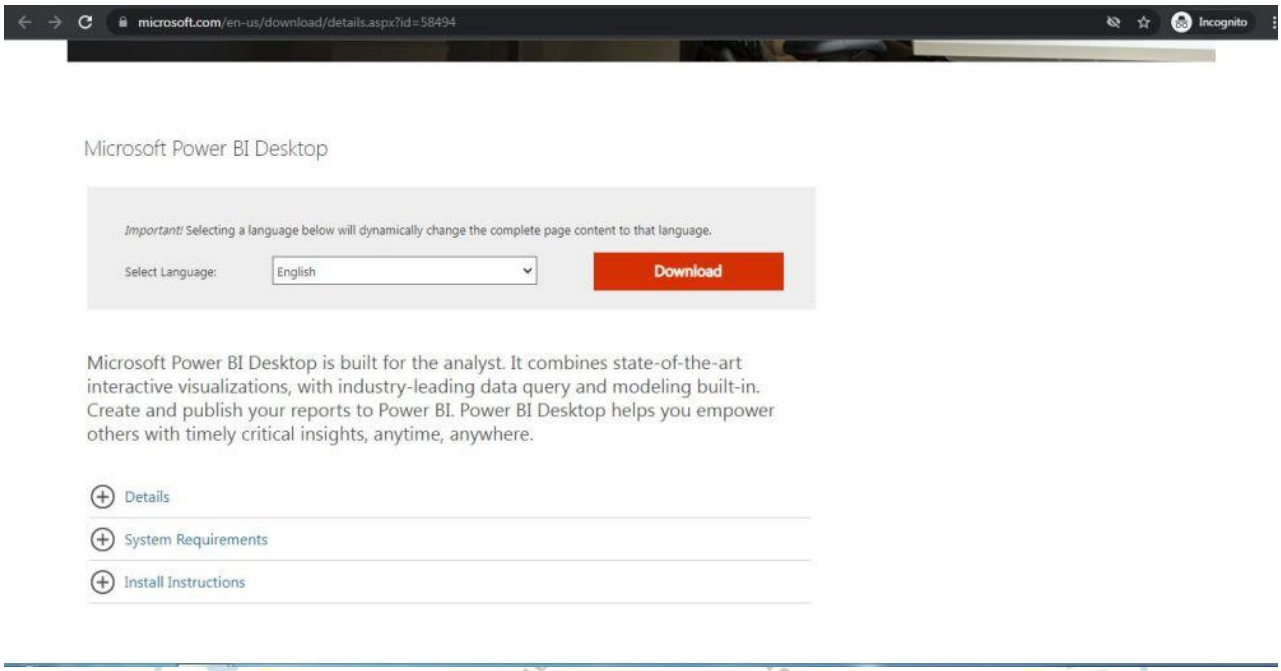
Step 2: Go to the Power BI Desktop Section

On the homepage click on the "Download Power BI Desktop" option.



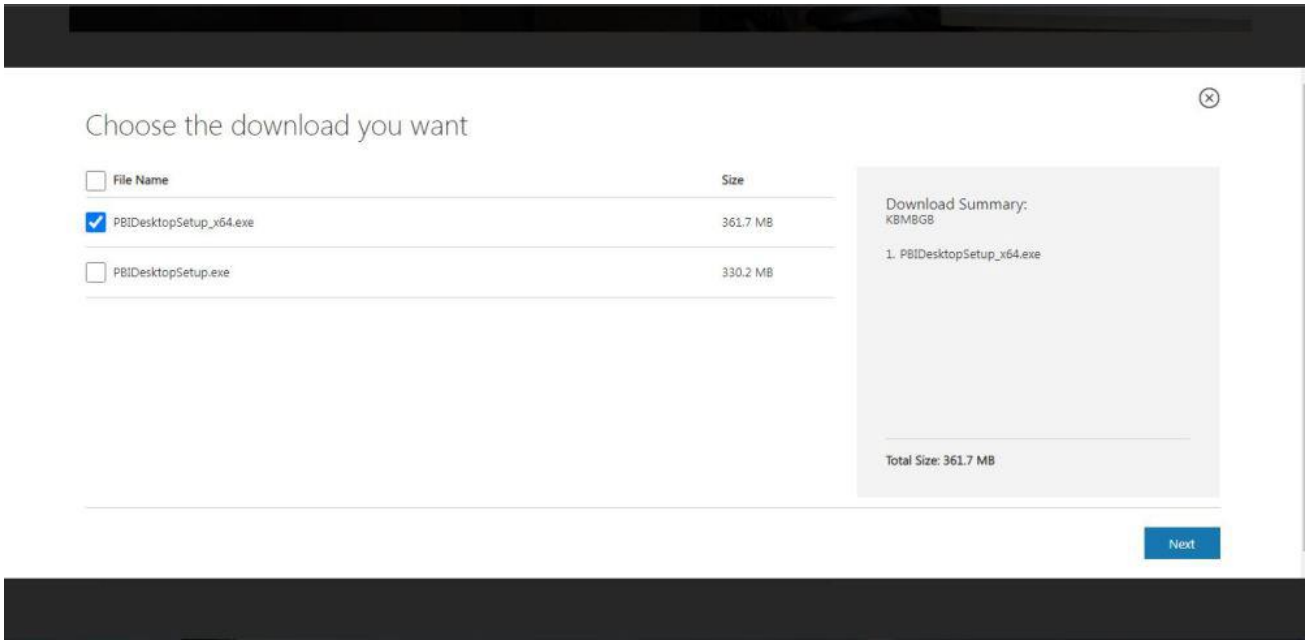
Step 3: Click on the Download Button

You will be taken to a new page. Click on the Download button to start the process.



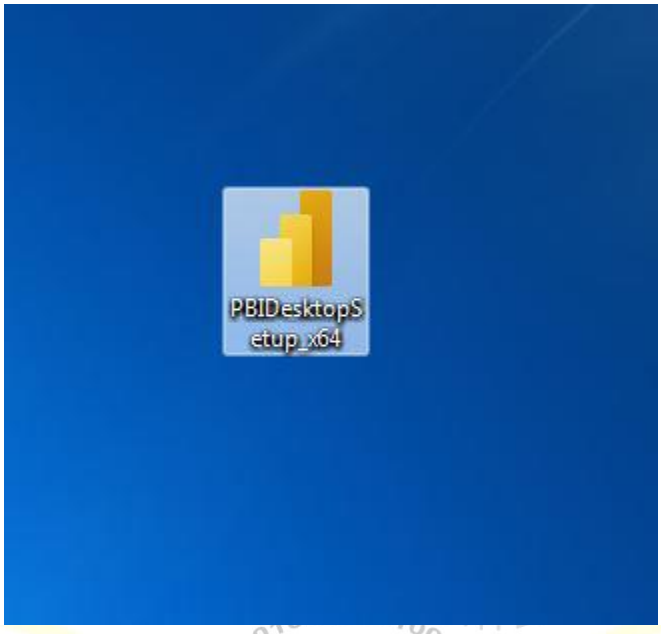
Step 4: Choose the Installer Based on Your System

On the next webpage choose the setup option according to your system configuration, let's take the first setup click on the Next button. Downloading of the executable file will start shortly. It is a big 361.7 MB file that will take some minutes.



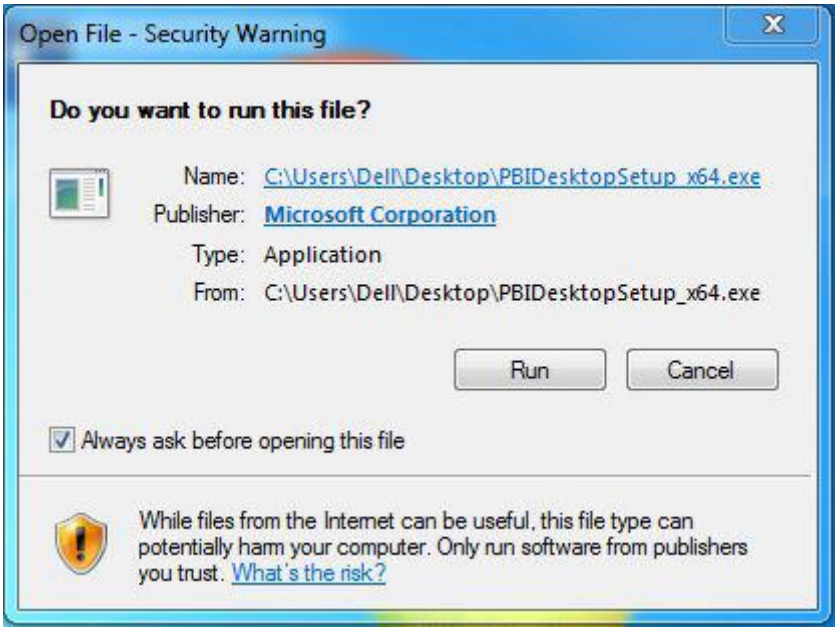
Step 5: Find the Downloaded File

Now check for the executable file in downloads in your system and run it.



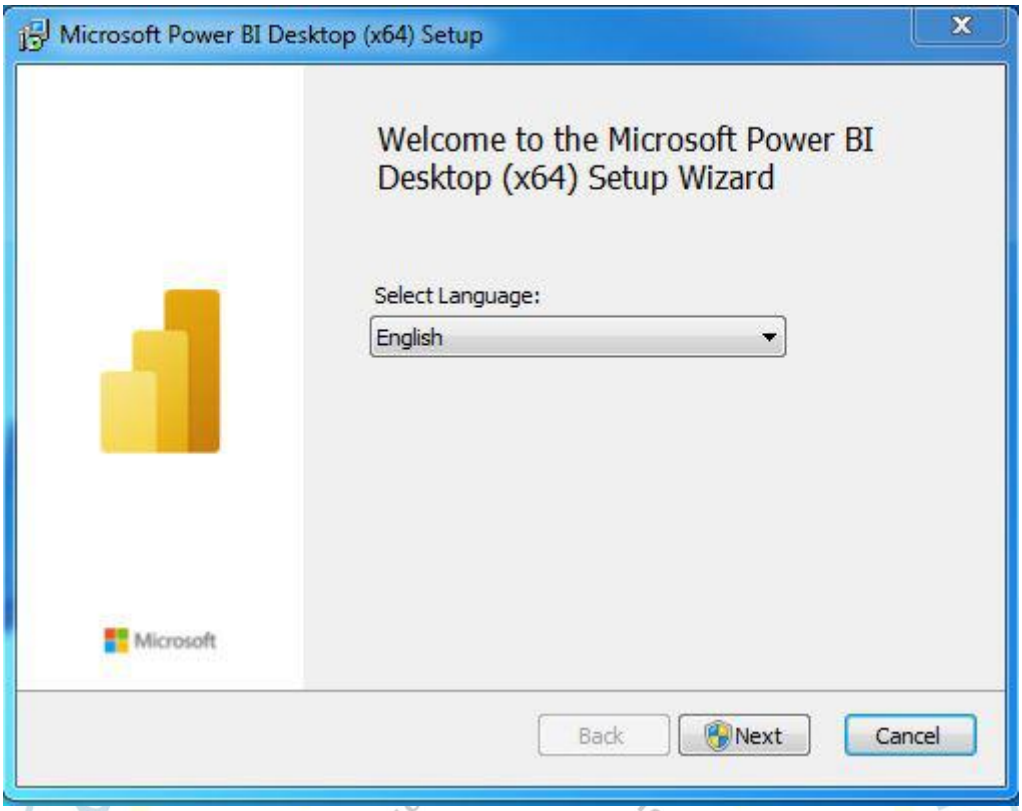
Step 6: Run the Installer

Double-click on the file. If prompted with a security warning click Run.



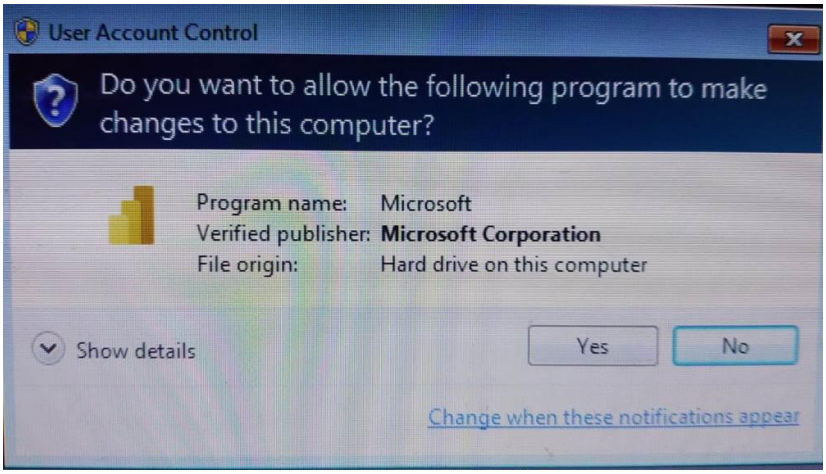
Step 7: Select Your Language

Choose your preferred language and click Next.



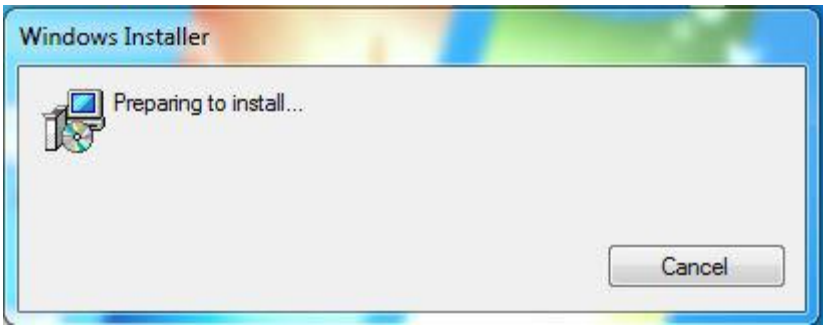
Step 8: Allow Changes to Your System

A pop-up will ask for permission to make changes to your system. Click Yes.



Step 9: Setup Will Start

Power BI will now begin preparing for the installation.



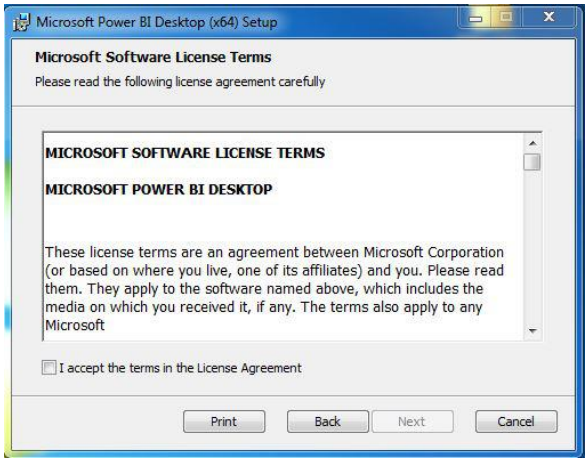
Step 10: Start Installation Setup

Click Next on the setup welcome screen.



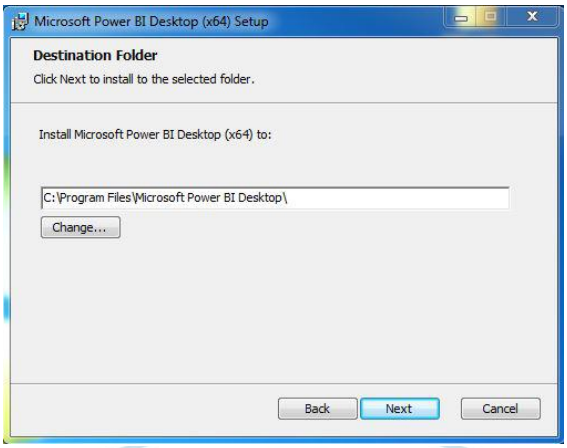
Step 11: Accept the License Agreement

Check the box that says “I accept the terms in the License Agreement” and click Next.



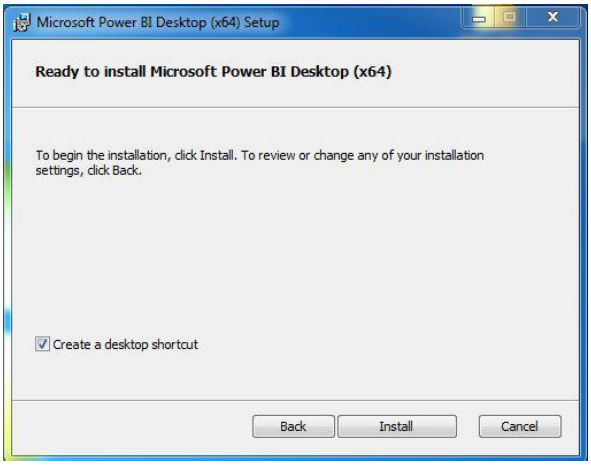
Step 12: Choose Installation Folder

You can either keep the default location or choose a different one. Then click Next.

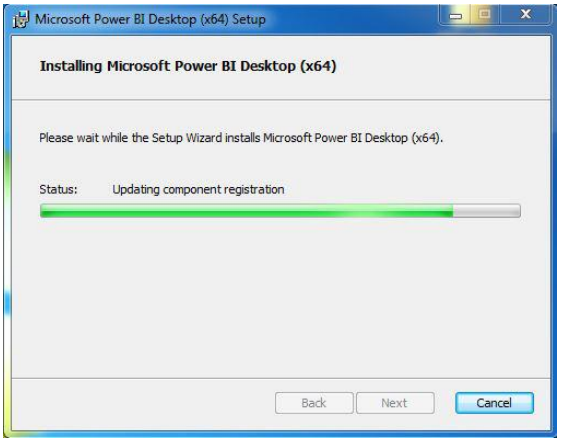


Step 13: Begin Installation

Click on the Install button. Installation will start and usually takes a minute or two.

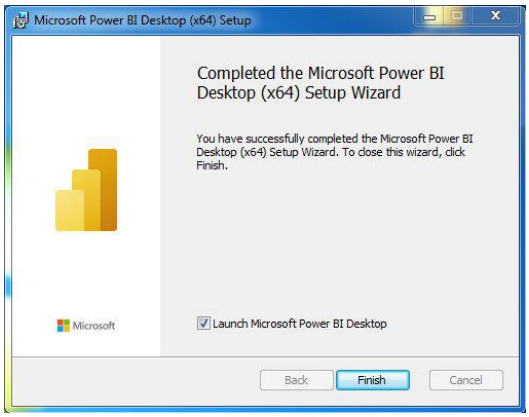


As you can see the installation is started in the below image



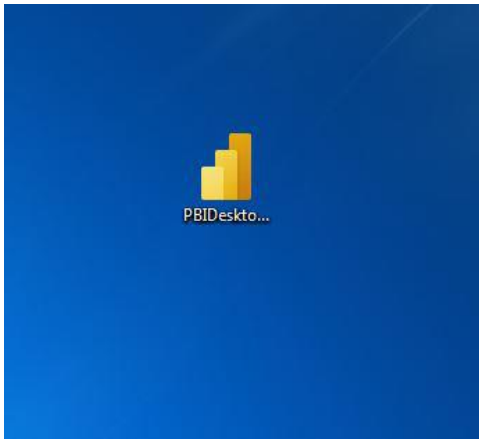
Step 14: Complete Installation

Once done click on the Finish button.

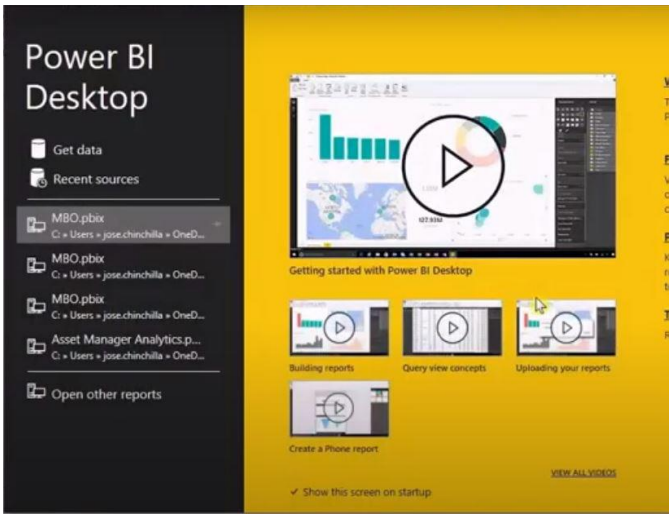


Step 15: Launch Power BI

A shortcut icon will appear on your desktop. Double-click it to open Power BI Desktop.



Now we run the software and see the interface.



Congratulations! At this point you have successfully installed Power BI on your windows system. You can now start exploring your data and building reports in easy manner.

POWER BI DESKTOP – PRACTICAL RECORD

Practical 1: Import Excel and CSV Files into Power BI Desktop

Objective: Import structured data from Excel and CSV sources.

Procedure:

1. Open Power BI Desktop.
2. Select Home > Get data > Excel. Choose PowerBI_Practical_Dataset.xlsx and select the Sales, Customers and Products sheets. Select Load (or Transform Data if you want to preview the queries).
3. Select Home > Get data > Text/CSV and open Sales_Messy.csv. Select Transform Data.
4. Check the Navigator/Power Query preview and then use Close & Apply.
5. Confirm the tables appear in the Data/Model panes.

Expected Result: Sales, Customers, Products and Sales_Messy tables are available in the report.

SCREENSHOT 1 – Get Data / Navigator and the loaded tables

Paste your Power BI Desktop screenshot here after completing the practical.

Practical 2: Clean Data by Removing Duplicates and Handling Missing Values

Objective: Clean an imported dataset before analysis.

Procedure:

1. Open Home > Transform data.
2. Select Sales_Messy.
3. Select the OrderID column and use Home > Remove Rows > Remove Duplicates.
4. Use the filter dropdown on Region, Product and UnitPrice to identify null values.
5. For Region and Product, replace nulls with an appropriate value such as 'Unknown' using Transform > Replace Values, or remove incomplete rows if the practical requires deletion.
6. For numeric UnitPrice, replace missing values using a justified value such as the median/appropriate business value. Record the rule used.
7. Select Close & Apply.

Expected Result: The cleaned table contains unique records and no unintended blank values.

SCREENSHOT 2 – Power Query Editor showing Remove Duplicates and null-value cleaning

Paste your Power BI Desktop screenshot here after completing the practical.

Practical 3: Change Data Types, Rename Columns and Split Columns

Objective: Prepare fields with correct types and useful names.

Procedure:

1. Open Transform Data.

2. In Sales, set Date to Date, Quantity to Whole Number, UnitPrice and Amount to Decimal Number.
3. Right-click UnitPrice and rename it to Unit Price.
4. Select a text column such as Product and choose Transform > Split Column > By Delimiter if a combined field is available.
5. If needed for demonstration, create a temporary combined field with a delimiter and then split it into two columns.
6. Close & Apply.

Expected Result: Columns have correct data types, clear names, and split fields where required.

SCREENSHOT 3 – Power Query Editor with data type icons, renamed column and Split Column menu
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 4: Create Custom Columns and Transform Data Using Power Query

Objective: Create a calculated field during data preparation.
Procedure:

1. Open Transform Data and select Sales.
2. Choose Add Column > Custom Column.
3. Use the formula: [Quantity] * [UnitPrice]
4. Name the column Calculated Amount.
5. Optionally add another custom column such as Sales Band using conditional logic.
6. Select Close & Apply.

Expected Result: A new Power Query column is created before the data reaches the model.

SCREENSHOT 4 – Custom Column dialog and the resulting Calculated Amount column
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 5: Create Basic Bar, Column, Line and Pie Charts

Objective: Build fundamental Power BI visuals.
Procedure:

1. Create a Clustered Column Chart: Axis = Product; Values = Sum of Amount.
2. Create a Bar Chart: Y-axis = Region; X-axis = Sum of Amount.
3. Create a Line Chart: X-axis = Date; Y-axis = Sum of Amount.
4. Create a Pie Chart: Legend = Category; Values = Sum of Amount.
5. Resize and arrange the four visuals on the report canvas.

Expected Result: Four basic visuals display sales information from the sample data.

SCREENSHOT 5 – Report canvas containing bar, column, line and pie charts
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 6: Format and Customize Visualizations

Objective: Improve readability and presentation of visuals.
Procedure:

1. Select a visual and open the Format visual pane.
2. Set a meaningful title, adjust data labels, axis titles, legend position and display units.
3. Apply a consistent theme from View > Themes.
4. Use the Format pane to change fonts, borders, background and other supported visual properties.
5. Repeat for the main visuals and align them neatly.

Expected Result: Visuals are consistently formatted with readable titles, labels and a coherent theme.

SCREENSHOT 6 – Format visual pane and customized dashboard
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 7: Create Relationships Between Multiple Tables

Objective: Build a relational data model.
Procedure:

1. Open Model view.
2. Drag Customers[CustomerID] to Sales[CustomerID].
3. Drag Products[ProductID] to Sales[ProductID].
4. Verify that the dimension tables are on the '1' side and Sales is on the '*' side where appropriate.
5. Check relationship direction and cardinality in the relationship dialog.
6. Return to Report view and test a visual using fields from related tables.

Expected Result: Customers and Products are related to Sales and fields can be used together in visuals.

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EENSHOT 7 – Model view showing Customers → Sales and Products → Sales relationships
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 8: Merge Two Related Tables Using a Common Key

Objective: Combine columns from related queries using Power Query Merge.
Procedure:

1. Open Transform Data.
2. Select Sales.
3. Choose Home > Merge Queries > Merge Queries as New.
4. Select Customers as the second table.

5. Select CustomerID in both tables as the matching key.
6. Use Left Outer join and select OK.
7. Expand the merged Customers column and choose CustomerName, City or Segment.
8. Close & Apply.

Expected Result: A new query contains Sales information enriched with selected Customer fields.

SCREENSHOT 8 – Merge Queries dialog with CustomerID selected in both tables
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 9: Append Multiple Datasets into a Single Table

Objective: Stack datasets with compatible column structures.
Procedure:

1. Open Transform Data.
2. Use Home > Append Queries > Append Queries as New.
3. Select two or more tables with the same column structure.
4. Choose Three or more tables when applicable and select the required queries.
5. Select OK and review the combined rows.
6. Rename the result ConsolidatedSales and select Close & Apply.

Expected Result: Rows from the selected datasets are combined into one consolidated table.

SCREENSHOT 9 – Append Queries dialog and consolidated table
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 10: Create Calculated Columns Using DAX

Objective: Create row-level calculations in the data model.
Procedure:

1. In Data or Model view, select the Sales table.
2. Choose Modeling > New column.
3. Enter: Sales Amount = Sales[Quantity] * Sales[UnitPrice]
4. Press Enter and verify the new column.
5. Optionally create: Profit = Sales[Sales Amount] - (Sales[Quantity] * RELATED(Products[Cost])) if the Products relationship is active.

Expected Result: The model contains DAX calculated columns that calculate values row by row.

SCREENSHOT 10 – DAX formula bar showing the Sales Amount calculated column
Paste your Power BI Desktop screenshot here after completing the practical.

Practical 11: Create Measures for Total, Average and Count

Objective: Create reusable DAX measures for aggregation.

Procedure:

1. Select Sales > New measure.
2. Create: Total Sales = SUM(Sales[Sales Amount])
3. Create: Average Sales = AVERAGE(Sales[Sales Amount])
4. Create: Order Count = DISTINCTCOUNT(Sales[OrderID])
5. Create a Card visual for each measure.
6. Test the measures with Region/Product fields to observe filter context.

Expected Result: Cards display total sales, average sales and distinct order count.

SCREENSHOT 11 – Three Card visuals showing the DAX measures

Paste your Power BI Desktop screenshot here after completing the practical.

Practical 12: Apply Slicers and Filters for Interactive Reports

Objective: Make the report interactive.

Procedure:

1. Insert a Slicer visual and add Region.
2. Add another slicer for Category or Product.
3. Use the Filters pane to add a visual-level or page-level filter.
4. Select a region in the slicer and observe all compatible visuals update.
5. Use Edit interactions if a visual should not respond to a particular slicer.

Expected Result: Users can select slicer values and dynamically filter the report.

SCREENSHOT 12 – Report page showing Region and Category slicers affecting multiple visuals

Paste your Power BI Desktop screenshot here after completing the practical.

Practical 13: Implement Drill-Down and Drill-Through

Objective: Navigate from summary information to detailed information.

Procedure:

1. For drill-down, create a hierarchy such as Year > Month > Date or Category > Product.
2. Place the hierarchy on a chart axis and enable the drill-down button on the visual.
3. Click a higher-level category to move to the next level.
4. For drill-through, create a new report page named Product Details.
5. Place Product in the Drill-through field well on that page.
6. Add a table/card with Product, Customer, Region, Quantity and Sales Amount.
7. From the main report, right-click a Product data point and choose Drill through > Product Details.

Expected Result: Drill-down reveals lower-level detail and drill-through opens a dedicated detail page.

SCREENSHOT 13 – Main report with drill-down controls and Product Details drill-through page

Paste your Power BI Desktop screenshot here after completing the practical

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Practical 14: Design an Interactive Dashboard and Prepare for Power BI Service

Objective: Combine multiple visuals into a single interactive report page and publish it.
Procedure:

1. Create a report page containing KPI Cards for Total Sales, Average Sales and Order Count.
2. Add a sales-by-region bar chart, product/category chart, sales trend line chart, and slicers.
3. Align and resize visuals using the Format/Arrange options.
4. Add a clear report title such as 'Sales Analysis Dashboard'.
5. Save the file as PowerBI_Practical_Report.pbix.
6. To publish, sign in to Power BI Desktop, select Home > Publish, choose an available workspace, and confirm.
7. After publishing, open the report in the Power BI Service and verify the report/semantic model. Sharing is subject to your organization's Power BI permissions/licensing.

Expected Result: A single interactive report page is ready and can be published to the Power BI Service when an appropriate account/workspace is available.

SCREENSHOT 14 – Completed interactive dashboard in Power BI Desktop and the Publish dialog
Paste your Power BI Desktop screenshot here after completing the practical.